

Orchestrating Microbiome Analysis with Bioconductor

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Abstract

The expansion of microbiome research has led to the accumulation of interlinked datasets encompassing versatile taxonomic and functional assays. While critical to advance the field, the analysis of increasingly large and heterogeneous multi-modal microbiome data would benefit from unified approaches supporting interoperability between methods and the design of modular data science workflows. The Bioconductor project has recently developed an optimized statistical programming framework for multi-assay data integration in computational life sciences. Building on this foundation, we introduce a community-developed open-source ecosystem for microbiome data science, dedicated to supporting joint analysis of hierarchical, interlinked, and heterogeneous datasets that are increasingly common in modern microbiome research. This interoperable infrastructure comprises of open-source datasets, methods, tutorials and an active online community. Its functionality and usage are detailed in the OMA book (<https://microbiome.github.io/OMA/docs/devel>), which offers guidance for prospective users and contributors.

Keywords: microbiome, Bioconductor, data science, data integration

Introduction

Modern data science applications critically rely on open-source methods created by the research community^{1;2} and open software ecosystems have become central innovation hubs, driving technological and cultural transformation³. The Bioconductor project has been established as a major distribution channel for open data science methods in the life sciences. Nowadays, its vast developer community maintains over 2,300 quality-controlled R packages for various fields of life science informatics⁴. Additionally, Bioconductor has evolved into a global community that supports data science collaboration and training⁵ in general, and now the international calls for enhancing microbiology education in particular⁶.

Microbiome research focuses on the study of microbial communities and their diversity, function, and interactions with the host and environment⁷. The technologies used in this field generate vast amounts of high-dimensional, hierarchically structured data. The expansion of microbiome research has led to the accumulation of interlinked, multi-modal datasets encompassing versatile taxonomic and functional assays. Microbiome data scientists are thus increasingly facing the need to integrate heterogeneous data across multiple modalities while implementing reproducible workflows, which presents many practical challenges. First, the unique statistical properties and complexity of microbiome data necessitate specialized approaches⁷⁻⁹. Secondly, the diversity of proposed solutions can make it difficult to identify the optimal methods and build interoperable workflows^{10;11}. Thus, the need for standardized microbiome data science infrastructures has been widely recognized⁷ and a variety of open microbiome data science frameworks have been developed, including Anvi'o¹², Mothur¹³, QIIME2¹⁴,

and phyloseq¹⁵. However, platforms that exclusively focus on microbiome data science may not be able to optimally address the need to link data across different multiple modalities originating from different research domains.

The Bioconductor project embeds the methods of microbiome data science into a wider statistical programming framework alongside with the concurrent development of interoperable methods and workflows for transcriptomics¹⁶, single-cell¹⁷, metabolomics¹⁸, proteomics^{18;19}, and other fields. Here, we introduce a community-developed open-source framework, supporting multi-modal data integration in microbiome research. We have extensively tested and documented the framework to facilitate the dissemination of evidence-based best practices in microbiome data science.

Data infrastructure

Data scientists routinely deal with interlinked assays, side information, and hierarchical data such as ontologies, trees, or networks. The ability to integrate heterogeneous data elements is essential in biological research, where the individual components of a system can be seldom studied in isolation. The challenge for a data scientist is that different methods often require variable and inconsistent input data formats, leading to technical complications and time lost on repeated data wrangling during routine analyses. Bioconductor is renowned for its ecosystem of interoperable statistical methods for life sciences. A key to its broad adoption has been its ability to mitigate those issues by providing standardized data structures supported by an ecosystem of interoperable methods. This helps to avoid wrangling data and aligning methods and leaves more time to be invested in the core analytical tasks.

Bioconductor data structures support statistical analysis of complex data combinations²⁰ and are widely adopted by various user communities. They have been implemented through specialized object-oriented classes that integrate multiple data elements into a single structure known as a *data container*²¹. The SummarizedExperiment (SE) class represents the primary data container underlying the Bioconductor ecosystem^{20;22}. Its position among the top-1% most downloaded packages in Bioconductor reflects its widespread recognition and increasing adoption among developers (Figure 1). SE provides a standardized solution to store tabular data by linking numeric abundance matrices with side information on features (rows) and samples (columns). Side information can include secondary details on the experiment, such as sample origin, collection time, or health status.

The interoperability of SE with other Bioconductor data structures facilitates the transfer of methods across different fields of omics research. This represents a major strength, which has facilitated the development of modular and scalable data science methods⁵. The shared data structures support the transfer of methods between omics fields, yielding improved overall quality control and user support. Consequently, the SE-based framework has been extended to similar data integration tasks in fields, such as single-cell and spatial omics through the SingleCellExperiment (SCE)¹⁷ and SpatialExperiment¹⁶ classes, respectively.

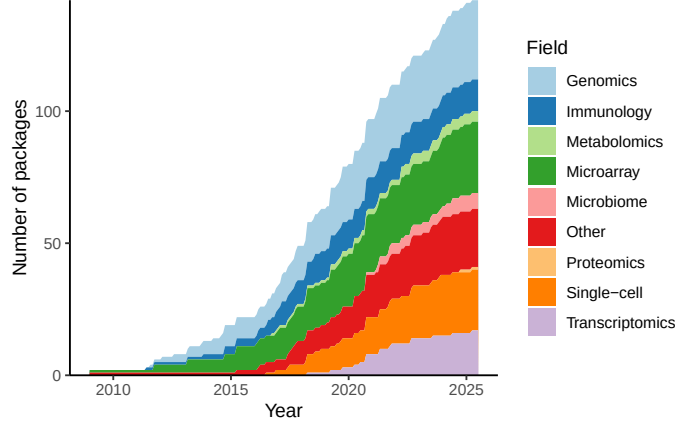


Figure 1: Adoption of the SummarizedExperiment (SE) data container among Bioconductor developers. The growth in the number of Bioconductor packages supporting SE over time is shown, categorized by their respective fields. For some of the packages SE support has been added after their original release.

Hierarchical data structures. Microbiome data scientists frequently need to integrate information on feature phylogenies and sample hierarchies in order to accurately reflect fine-scale variability in microbiome data. While maintaining interoperability with the broader Bioconductor ecosystem, the `TreeSummarizedExperiment` (`TreeSE`) class²³ extends the SE and SCE data structures to hierarchical datasets by incorporating feature and sample organisations as tree structures (Figure 2). Moreover, `TreeSE` supports the storage of other common aspects of microbiome data, such as reference sequences, and incorporates results from common analytical procedures such as dimensionality reduction, statistical transformations, and agglomeration by taxonomic, functional or other information. The interoperability with other Bioconductor classes facilitates scalable analysis and the design of modular yet flexible workflows. The direct interoperability with the widely adopted SE data science ecosystem distinguishes our `TreeSE`-based approach from `phyloseq`, which is another popular Bioconductor data container originally developed with a focus on 16S amplicon data analysis¹⁵. `TreeSE` is a superset that can accommodate all elements of a `phyloseq` object, while also offering improved support for multi-modal data and often substantial gains in memory efficiency and computing speed (Figure 3). Overall, the `TreeSE` approach outperforms `phyloseq` in these benchmarks. While `speedyseq` is faster in certain operations, `TreeSE` scales more efficiently in both execution time (t) and allocated memory (m) as sample size (n) increases. Conversion functions and detailed cheat sheets between the two formats are available to bridge the gap between the respective user communities.

Multi-assay data structures. When integrating data across multiple modalities, such as taxonomy, metabolomics, or host genomics, more flexible sample mapping may become necessary. Multi-omics data is often sparse, and samples can be only partially matched between experiments. In other cases, a single sample in one modality can

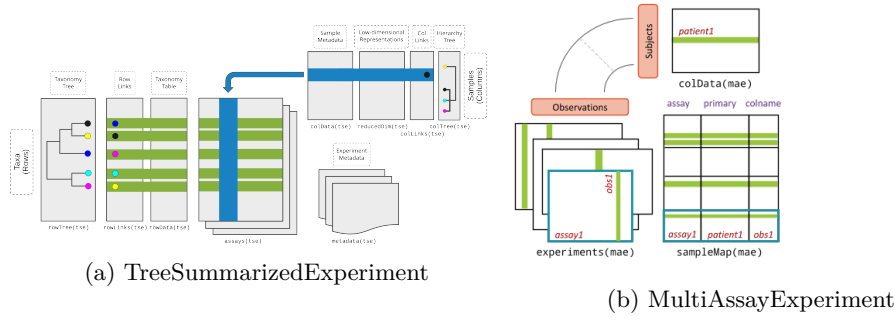


Figure 2: Bioconductor data containers for microbiome data. **(a) TreeSummarizedExperiment (TreeSE).** TreeSE supports hierarchical and multi-table analysis in microbiome data science²³. It links the numeric abundance matrix and potential alternative versions (statistical transformations, different taxonomic levels, dimensionality reduction) with tabular and hierarchical side information on the features (rows) and samples (columns). Additional slots for reference sequences and experiment metadata are available. (Figure modified from²³) **(b) MultiAssayExperiment (MAE).** MAE facilitates integration of data from multiple experiments²⁴. (Figure from²⁴)

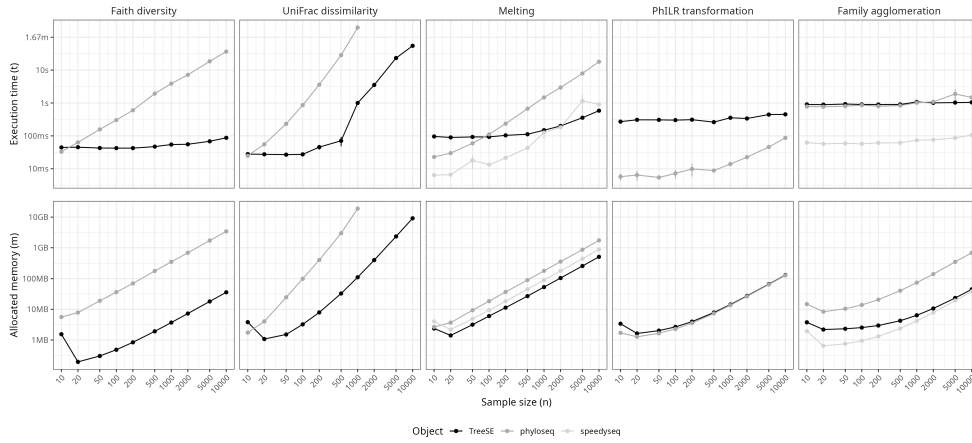


Figure 3: Comparison of execution time and memory consumption between alternative methods in Bioconductor. TreeSE, phyloseq and speedyseq were benchmarked for five common analytical operations applied to random subsets of samples from a large study on wild baboons²⁵ (accessible through the microbiomeDataSets Bioconductor package). The source code for the benchmarking experiments can be accessed through the OMA online book.

be matched by two or more samples in another modality see e.g.²⁶. The MultiAssayExperiment (MAE) data container²⁴ facilitates the integration of multiple data objects representing different omics modalities (see Figure 2). Whereas TreeSE is restricted to a one-to-one sample mapping between different data tables, MAE supports flexible mappings between multiple experiments from the same study. Integration of multi-table data without such a structured framework often leads to ambiguous mappings and inconsistent results. For instance, diagonal integration methods^{27;28} – where shared features across data types may not be explicitly defined – can benefit substantially from such a systematic framework that enforces anchor alignment. This mirrors lessons from single-cell data integration, where shared anchors and hierarchical mapping have proven essential for interpretability. Flexible linking of samples across several data objects supports efficient data integration, methods sharing, and overall interoperability.

Table 1 summarizes the microbiome-related data containers, while Table 2 introduces the available data elements in each container.

Table 1: Recommended data containers for microbiome analysis.

Data container	Target use
SummarizedExperiment (SE) ²⁰	generic data container for experiments with multiple assays
TreeSummarizedExperiment (TreeSE) ²³	tailored data container for hierarchically organized experiments
MultiAssayExperiment (MAE) ²⁴	tailored data container for sets of multi-omic experiments

Table 2: Key data elements in the employed data containers.

Slot	Content	SE	TreeSE	MAE ^a
Tables				
assays	tables of features (rows) and samples (cols)	X	X	X
altExps ^b	experiments with variable number of features		X	X
reduced-Dims	results of dimensionality reduction		X	X
Rows				
rowData	side information on features	X	X	X
rowPairs	linkage between pairs of associated features		X	X
rowTrees	hierarchical organization of features		X	X
referenceSeq	reference sequences of features		X	X
Columns				
colData	side information on samples	X	X	X
colTrees	hierarchical organization of samples		X	X
sampleMap ^c	flexible sample mapping across experiments			X

Slot	Content	SE	TreeSE	MAE ^a
Other				
metadata	additional information on the experiment	X	X	X

^a While MAE does not contain conventional slots (with the exception of colData), its experiments can include any number of data containers that provide those slots. ^b altExp samples must match assay samples in a 1-to-1 mapping. ^c sampleMap supports diverse mappings between samples from different experiments (1-to-1, 1-to-many, many-to-1 and many-to-many).

Data resources. Several open microbiome data resources are supported, enabling direct data import into the aforementioned data containers for further downstream analysis within the Bioconductor methods ecosystem (see Table 3). Examples of the available data resources include curatedMetagenomicData²⁹, HoloFood³⁰, microbiomeDataSets³¹, and the EBI/MGnify database³². In addition, Bioconductor packages often include built-in demonstration datasets. Collectively, these resources encompass taxonomic and functional profiles from thousands of published microbiome studies and tens of thousands samples that can be accessed with the available tools for research and educational purposes.

Table 3: Open microbiome data resources supporting the framework.

Data resource	Package	# studies or datasets	# samples
Curated metagenomic data ²⁹	curatedMetagenomicData ²⁹	93	22,588
HoloFood ³⁰	HoloFoodR ³³	-	9,990
MGnify ³²	MGnifyR ³⁴	5129	616,138
microbiomeDataSets ³¹	microbiomeDataSets ³¹	6	19,100

Data preparation

While microbiome data scientists continue to develop specialized solutions^{8;9;35}, the implementations typically differ in their expected input and output formats. Common data standards support modular data science workflows, speeding up methods development, benchmarking, and replication. Here, we describe an ecosystem of software packages that has recently emerged to support microbiome data integration and analysis based on the data containers, (Tree)SE and MAE (see Figure 4).

The framework focuses primarily on the downstream analysis of high-throughput sequencing data from microbiome experiments after the original measurements have already been summarized into abundance tables representing e.g. taxonomic or functional profiles. Taxonomic abundance tables are one of the most commonly encountered data types. Common measurement platforms include amplicon-based marker gene

studies (e.g. 16S rRNA) and metagenomic sequencing, but alternative profiling techniques are available including phylogenetic microarrays³⁶ and high-density optical mapping³⁷. Microbiome analyses can also yield variable metagenomic, and functional features. The interoperability of the data formats supports modular workflow design for integrating different omics through the Bioconductor ecosystem. This decisively mitigates reproducibility challenges posed by rapidly evolving reference databases and profiling tools (e.g., MetaPhlAn2 vs. MetaPhlAn4) by enabling structured coexistence and direct comparison within a unified, version-resilient framework^{38;39}.

Data import. Most users will need to import data for their own analyses. The TreeSE data container can be populated with microbiome data from common tabular data types but non-standard file formats may require additional data wrangling. To streamline data import, we provide importers for a variety of standard microbiome file formats including BIOM⁴⁰, MetaPhlAn/HUMAnN (MetaPhlAn v. 2-4, HUMAnN v. 3)⁴¹, Mothur¹³, QIIME2¹⁴ and the TAXonomic Profile Aggregation and STandardisation format (taxpasta)⁴². These importers can automatically load the data into the appropriate data containers, thus forming a bridge between low-level sequence processing methods and the high-level statistical programming framework in R. Moreover, converters between alternative data structures in R are available supporting seamless conversions between TreeSE and BIOM, DADA2⁴³ and phyloseq (the `mia::convert*` functions), and TreeSE-based microbiome data objects can be exported into external services such as MicrobiomeDB⁴⁴ and in other languages (e.g. Julia). These bridges facilitate the broadest possible access to microbiome analysis methods.

Quality control and exploration. After importing the data into R, the recommended first step involves basic data exploration and quality control⁴⁵. In addition to methods for identifying contaminant sequences and calculating summary statistics, the framework includes a versatile set of custom visualization methods to assess data variability, outliers, homogeneity of variance and other aspects that may need to be considered in downstream data analysis. The `mia` and `miaViz` packages, for instance, provide convenient access to many operations for common exploratory tasks in microbiome analysis and visualization for TreeSE objects (Figure 5).

Filtering and agglomeration. Microbiome datasets may contain tens of thousands of unique features depending on the sequencing resolution, databases, and the scope of analysis. Often – to summarize data into biologically meaningful groups or to reduce multiple comparisons – the data is subsetting, filtered or agglomerated to higher-level taxonomic or functional categories or stratified by sample groups. The package ecosystem and the `mia` package feature various tools for such tasks. For instance, the prevalence methods can be useful to remove very rare features that might be sequencing artifacts or present otherwise limited value for the analysis. The mechanism of alternative experiments (`altExp`) supports data agglomeration by taxonomic levels and by other features such as pathogenicity or functional potential (e.g. butyrate production⁴⁶, gut-brain modules⁴⁷, and other host-associated signatures⁴⁸). The supporting methods incorporate the agglomerated levels within the same data object for parallel operations, and typically also take care of agglomerating the associated tree information and the alternative assays. The filtering, subsetting, and agglomeration operations

are thus supported in both feature as well as sample spaces, and simultaneously propagate to cover the relevant alternative assays, alternative experiments, ordinations, and side information. This supports the efficient management of interlinked assays and saves time for dedicated expert tasks.

Transformation. Common statistical transformations in microbiome research include relative abundance⁴⁹, centered log-ratio (CLR) and its robust version (rCLR), which are commonly used to mitigate compositionality bias⁵⁰, and the tree-aware phylogenetic ILR (phILR) transformation⁵¹. Data rarification is a related concept that has been suggested to control for uneven sequencing depths and it remains widely debated in microbial ecology^{52;53}. Our framework supports these and a variety of other transformations commonly applied to microbiome data.

Data enrichment. Finally, incorporating additional data is a frequent operation in applied data analysis, and it often takes place in an iterative fashion. The data containers support gradual and organized accumulation of interlinked data elements including transformations, aggregated data, dimension reduction, and derived data on samples and features. We generally recommend that the data objects are initialized at the beginning of the workflow, where they can then be gradually enriched as the project advances. By systematically extending the upstream data preparation step – instead of ad-hoc calculations scattered along the workflow – the data scientist can achieve well-organized and transparent provenance of the data elements. Data can be enriched by importing new external information as well as by deriving new information from the data itself into the same data container as described in the next section.

Downstream data analysis

The new Bioconductor framework for microbiome data science improves the management of integrative analysis workflows. It facilitates systematic linking of the different data elements and analysis outcomes, helping to shift the focus towards multi-modal analyses and methods development while supporting standard analysis tasks in microbiome research.

Microbiome data analysis often relies on specialized statistical approaches as data sparsity, zero-inflation, compositionality, and heteroschedasticity pose challenges for standard data analytical methods^{8;49}. Moreover, microbiome data is hierarchically structured across taxonomic, ecological, and spatial levels^{7;54}. Alternative processing pipelines pose different biases and generate varying data formats⁵⁵. These and other unique characteristics have wide-reaching implications for microbiome analysis, necessitating dedicated analysis methods⁷. Our statistical programming framework supports the community-driven development and benchmarking of the solutions. The following sections provide an overview of the available tools.

Community indicators. Various global indicators are available to quantify the microbial community state or other sample properties. Community diversity represents a common measure in microbial ecology⁵⁶ (Figure 5). Alpha diversity metrics generally quantify the observed or estimated number of distinct species (“richness”) and

their distribution (“evenness”). Many diversity indices, such as the Shannon index, combine these two aspects. Phylogenetic diversity measures additionally consider the phylogenetic relatedness of community members⁵⁷; the new implementation of the Stacked Faith’s Phylogenetic Diversity can accelerate the performance 1-2 orders of magnitude⁵⁸. Enhanced analyses with repeated rarefaction are supported⁵⁹ albeit the use of rarefaction remains controversial^{53;60}. Other global indices of community state include measures of dominance, rarity, skewness, kurtosis, modality, library size (number of reads), or absolute abundance quantification. Sample information in the TreeSE object (colData) can be enriched with these derived quantities. By default, the `mia::addAlpha` method returns a set of complementary alpha diversity measures⁶¹.

Community similarity is commonly evaluated with beta diversity measures, which quantify multivariate dissimilarity and can be used to quantitatively compare community composition. The framework supports common indices in microbial ecology including Bray-Curtis, Jaccard, Aitchison^{50;56}, and the phylogeny-aware UniFrac⁶². The community (dis)similarities are commonly visualized with unsupervised ordination methods that project the data on a lower-dimensional display⁶³. Unsupervised techniques, such as Principal Component Analysis (PCA) and Principal Coordinates Analysis (PCoA) also known as Multi-Dimensional Scaling (MDS), and Non-metric MDS (NMDS) (Figure 5) are common choices, with enhanced support for microbiome analysis and visualization. Statistical tests (e.g. PERMANOVA) and supervised visualization methods, such as the distance-based redundancy analysis (dbRDA) can complement the unsupervised analyses, and help to uncover and quantify co-variation between community composition and covariates. This can benefit from compositionally robust measures (e.g. robust Aitchison distance⁵⁰), or phylogeny-aware dissimilarity measures specifically developed for microbiome research (e.g. UniFrac⁶²). Sample dissimilarities provide the basis for many specialized microbiome techniques such as microbial community typing (clustering) with Dirichlet Multinomial Mixtures (DMM)⁶⁴ and community state types (CST)⁶⁵, or factorization into community signatures⁶⁶. These and other multivariate methods are provided for instance by `mia`, `miaViz`, and additional single-cell packages (`scater`, `scuttle`¹⁷).

Differential abundance and mediation analysis. Differential abundance analysis (DAA) and differential prevalence analysis (DPA) methods are used to identify microbial taxa whose abundances vary across study groups or continuous gradients (Figure 5). Many dedicated DAA methods in microbiome research support the TreeSE framework including methods such as ANCOM-BC2⁶⁷, LimROTS¹⁹, MaAsLin3⁶⁸, and radEmu⁶⁹. Recent studies by us and others indicated that elementary methods, such as log-transformed TSS-counts coupled with a linear model, or presence/absence data with logistic regression often yield the most consistent results⁷⁰. The standard DAA typically focuses on individual features, ignoring interactions and potential co-abundance variation. Thus, we support co-abundance analyses; agglomerating collinear features as a preprocessing step can reduce multiple testing, increase statistical power, and support interpretation but is also prone to subjective parameterization and limited generalizability. The grouping can be based on clustering of the abundance profiles or on external information such as phylogenetic relations, taxonomic classifications or

functional properties (e.g. butyrate producers⁴⁶). Beyond these, advanced methods—such as causal mediation analysis tailored to microbiome multi-omics data—are already being developed using SummarizedExperiment⁷¹, highlighting its readiness to support emerging innovations in microbiome analytics.

Function- and sequence-level analyses. Taxonomic analysis is frequently complemented by functional analyses, either based on functional predictions derived from amplicon or metagenomic sequences, or from parallel metabolomic or other functional assays. Our framework not only supports incorporation of functional abundance tables within the same data objects but specialized methods on functional data are available; for instance, regarding metabolome analyses the Notame package⁷² recently added SE support, and the R4MS project provides many interoperable tools¹⁸. The framework has found recent applications in extended analyses of metagenomic features, i.e. antimicrobial resistance gene composition, diversity, and load⁷³. We have converted a large-scale compilation of resistome profiles in 8,972 metagenome samples⁷⁴ in the TreeSE format for research and educational purposes.

Spatio-temporal, ecological and epidemiological models. Microbiome time series⁷⁵ encompass a wide range of study designs, ranging from short to long follow-up times, sparse to dense data, univariate to multivariate and multi-omic monitoring⁷⁶, and real, pseudo, or simulated time series. The miaTime package supports longitudinal microbiome analyses. Time series from standard models of microbial community dynamics including e.g. the Hubbell’s neutral model, generalized Lotka-Volterra, and consumer-resource model can be simulated with the miaSim package⁷⁷. Moreover, an increasing number of microbiome studies are incorporating the spatial dimension⁵⁴. The compatibility between data structures support the adoption of solution from other application domains in Bioconductor, such as single-cell sequencing¹⁷ and spatial transcriptomics¹⁶ (Figure 5). Spatial and temporal information can be incorporated as a covariate or by estimating temporally or spatially resolved metrics, such as divergence, auto-correlation and modality. Another type of temporal analysis includes the time-to-event, or survival models, which are encountered in e.g. epidemiological and population cohort studies; specialized survival analysis strategies for microbiome data have been proposed^{78;79}.

Multi-modal integration. An increasing number of studies integrate microbial abundance data with further omics, such as transcriptomics, proteomics, and host genomics^{7;80–82} (Figure 5). As demonstrated above, Bioconductor data containers support combinations of complex omics data.^{17;20;24} Statistical methods for multi-table analysis in microbiome research can be broadly categorized into three groups: Machine Learning (ML) predictive modelling⁸³, association-based modelling, and latent variable modelling.^{7;8;83–86} Some of the currently supported multi-omic methods include cross-correlation analyses, canonical correspondence analysis (CCA), anansi⁸⁵, and joint-RPCA [in press].

Other methods and programming environments. A range of other methods are employed in microbiome data science. Examples include network analysis⁸⁷, ML-based prediction and classification⁸⁸, enrichment analyses such as BugSigDB⁴⁸ or Microbe

Set Enrichment Analysis (MSEA)⁸⁹. The recent adoption of interoperable data containers and the methodological similarities in related fields can greatly expand the selection of available toolkits in emerging areas. In addition to the internal coherence within the Bioconductor ecosystem, this framework supports integration with external platforms through interfaces such as `reticulate` for Python, `Rcpp` for C++, and compatibility with pipelines like `bioBakery`^{41;90}. These connections empower users to combine the best tools available across environments and languages while maintaining structured data interoperability via standard containers like `TreeSE` and `MAE`.

Accessible analysis and community

Modern science is embodied by a critical community capable of assessing discoveries and replicating results⁹¹. Open software and the social structures around it present a timely manifestation of this principle, with technical, legal, sociological and epistemic consequences³. FAIR principles for software support access to well-documented and user-friendly methods¹, and are facilitated by the extensive quality control and documentation standards enforced by Bioconductor²⁰. In addition to these technical demands, the development and quality control of academic software critically relies on active critical source communities that can provide continuous benchmarking and support the identification and dissemination of evidence-based best practices^{1;3;75}. We actively support community formation by online training resources and communication channels. These, together with the interactive applications, provide accessible entry points for prospective developers and users.

The OMA book. *Orchestrating Microbiome Analysis with Bioconductor (OMA)* (<https://microbiome.github.io/OMA/docs/devel/>) is an online book that provides a comprehensive introduction to the microbiome data science framework in Bioconductor. It offers practical guidance, reproducible workflows, and training material for integrative microbiome analyses. The book is versioned to match Bioconductor's biannual software release cycles. The work incorporates rich feedback from the user community and microbiome training courses. It complements the broader trend, where specific Bioconductor data structures^{20;23;24} have been adopted as a solution for multi-table data integration tasks in various domains such as single-cell sequencing (OSCA book)¹⁷ and transcriptomics *OSTA book*. These resources demonstrate how state-of-the-art data containers can support statistical programming on interlinked multi-table data sets in a highly optimized way.

Interactive applications. Graphical user interfaces provide accessible entry points to the framework. The `iSEETree` and `miaDash` packages provide interactive analysis and visualization tools for microbiome data⁹² (Figure 5). Their interface supports automatic generation of source code and replicable workflows as well as analysis templates for users without a strong programming expertise. Moreover, the tidy R paradigm, available through `tidyomics`, simplifies the syntax and can streamline the workflows⁹³. These tools allow users to perform basic analyses and visualize data with ease.

Community. Its widespread adoption has made R one of the most cited software resources of our time² and the introduction of the shared data structures facilitates the broad interoperability of the methods ecosystem. Bioconductor has a strong tradition in fostering the R user community through shared data structures and training activities⁵. An active community of developers and users thrives online, supporting the software sustainability and promoting good practices in scientific computing⁹⁴. The community interacts via several online communication channels, including Bioconductor Zulip, a Github forum, and an email list. The development of the framework is influenced by feedback from the user community through these channels, regular meetings and training workshops.

Outlook

The Bioconductor community has a well-established reputation in developing open, peer-reviewed software for life sciences. Its strengths include the enforcement of minimal documentation standards, semantic versioning, portability across operating platforms, quality control (e.g. unit testing, continuous integration), usability (APIs), and robustness (deprecation mechanisms), and a biannual release cycle. During the past two decades Bioconductor has grown into a repository of 2,300 actively maintained software packages, of which 58 are classified under the Microbiome Bioc task view. In order to enjoy the benefits of automation while maintaining flexibility, data scientists routinely remix open-source software into custom workflows. However, the lack of commonly adopted data representation standards often hinders the development, sharing, and application of statistical workflows. At the core of the Bioconductor’s data science strategy is the idea of shared data representation. The enhanced interoperability can help to minimize redundant efforts and facilitate the translation of methods between different research areas^{1;95} and, importantly, facilitates the formation of broad developer and user communities.

Microbiome research has shifted towards integrative analyses aiming at mechanistic and causal inference^{56;96}. Technological advances in long-read sequencing, functional assays, spatial profiling, and single-cell have created an increasing demand for integrative data analysis techniques. Thus, microbiome data is increasingly linked with other omics along varying spatial and temporal dimensions. Our framework addresses this need, together with parallel developments in other areas of omics.

Recent developments in microbiome and multi-omics workflows increasingly rely on Bayesian modeling and Gaussian processes to accommodate uncertainty and temporal or spatial structures⁹⁷. Our framework is compatible with these approaches, providing the structured input necessary for probabilistic inference and scalable computation. Building on these statistical foundations, the rapid emergence of foundation models and deep learning approaches in biomedicine calls for robust, scalable data representations⁹⁸. Our framework, through its use of standardized containers and modular workflows, is well-positioned to support such advances. Applications include transfer learning across studies, interpretable embeddings of microbiome multi-omic profiles, computational

prediction of microbial function and metabolites^{99;100}, and integration with AI agents for hypothesis generation and automated analysis, among others^{101;102}.

Open data science infrastructures have become critical components of reproducible research and a vast body of applications critically rely on open-source methods and workflows^{1;3;75}. Standardized data structures can enhance the transparency, reliability, and explainability of analysis workflows. We have documented the interoperable framework built on widely adopted data containers, open data resources, analysis methods and reproducible workflows for multi-modal microbiome research. Further developments could include extended methodology for temporal, spatial, and multi-omic analyses. Supporting interoperability between R and other programming languages can facilitate the application of other ML frameworks. As the field evolves, Bioconductor will continue to integrate new tools and approaches to improve integrative analyses of microbiome data.

References

- [1] Barker, M. *et al.* Introducing the FAIR principles for research software. *Scientific Data* **9**, 622 (2022).
- [2] Pearson, H., Ledford, H., Hutson, M. & Van Noorden, R. The most-cited papers of the twenty-first century. *Nature* **640**, 588–593 (2025).
- [3] Hocquet, A. *et al.* Software in science is ubiquitous yet overlooked. *Nature Computational Science* **4**, 465–468 (2024).
- [4] Gentleman, R. *et al.* Bioconductor: Open software development for computational biology and bioinformatics. *Genome Biology* **5**, R80 (2004).
- [5] Drnevich, J. *et al.* Learning and teaching biological data science in the Bioconductor community. *PLoS Computational Biology* **21**, e1012925 (2025).
- [6] Timmis, K. *et al.* A concept for international societally relevant microbiology education and microbiology knowledge promulgation in society. *Microbial Biotechnology* **17**, e14456 (2024).
- [7] Moreno-Indias, I. *et al.* Statistical and machine learning techniques in human microbiome studies: Contemporary challenges and solutions. *Frontiers in Microbiology* **12** (2021).
- [8] Jeganathan, P. & Holmes, S. P. A statistical perspective on the challenges in molecular microbial biology. *Journal of Agricultural, Biological and Environmental Statistics* **26**, 131–160 (2021).
- [9] Willis, A. D. Rigorous statistical methods for rigorous microbiome science. *mSystems* **4**, e00117–19 (2019).

- [10] Wen, T. *et al.* The best practice for microbiome analysis using R. *Protein & Cell* **14**, 713–725 (2023).
- [11] Shetty, S. A. & Lahti, L. Microbiome data science. *Journal of Biosciences* **44**, 115 (2019).
- [12] Eren, A. M. *et al.* Community-led, integrated, reproducible multi-omics with anvi'o. *Nature Microbiology* **6**, 3–6 (2021).
- [13] Schloss, P. D. *et al.* Introducing Mothur: Open-source, platform-independent, community-supported software for describing and comparing microbial communities. *Applied and Environmental Microbiology* **75**, 7537–7541 (2009).
- [14] Bolyen, E. *et al.* Reproducible, interactive, scalable and extensible microbiome data science using QIIME2. *Nature Biotechnology* **37**, 852–857 (2019).
- [15] McMurdie, P. J. & Holmes, S. phyloseq: a Bioconductor package for handling and analysis of high-throughput phylogenetic sequence data. *Pacific Symposium on Biocomputing* 235–246 (2012).
- [16] Righelli, D. *et al.* SpatialExperiment: infrastructure for spatially-resolved transcriptomics data in R using Bioconductor. *Bioinformatics* **38**, 3128–3131 (2022).
- [17] Amezquita, R. A. *et al.* Orchestrating single-cell analysis with Bioconductor. *Nature Methods* **17**, 137–145 (2020).
- [18] Gatto, L., Rainer, J., Soneson, C. & Nishida, K. *R for Mass Spectrometry* (2025). URL <https://zenodo.org/record/15180830>.
- [19] Anwar, A. M., Jeba, A., Lahti, L. & Coffey, E. LimROTS: A hybrid method integrating empirical Bayes and reproducibility-optimized statistics for robust analysis of proteomics data. *bioRxiv* (2025).
- [20] Huber, W. *et al.* Orchestrating high-throughput genomic analysis with Bioconductor. *Nature Methods* **12**, 115–121 (2015).
- [21] Chambers, J. M. Object-oriented programming, functional programming and R. *Statistical Science* **29** (2014).
- [22] Morgan, M., Obenchain, V., Hester, J. & Pagès, H. *SummarizedExperiment: A container (S4 class) for matrix-like assays* (2025). R package version 1.38.1.
- [23] Huang, R. *et al.* TreeSummarizedExperiment: a S4 class for data with hierarchical structure. *F1000Res* **9**, 1246 (2021).
- [24] Ramos, M. *et al.* Software for the integration of multiomics experiments in Bioconductor. *Cancer research* **77**, e39–e42 (2017).

- [25] Grieneisen, L. *et al.* Gut microbiome heritability is nearly universal but environmentally contingent. *Science* **373**, 181–186 (2021).
- [26] Husso, A. *et al.* Impacts of maternal microbiota and microbial metabolites on fetal intestine, brain, and placenta. *BMC Biology* **21**, 207 (2023).
- [27] Xu, Y. & McCord, R. P. Diagonal integration of multimodal single-cell data: potential pitfalls and paths forward. *Nature Communications* **13**, 3505 (2022).
- [28] Argelaguet, R., Cuomo, A. S., Stegle, O. & Marioni, J. C. Computational principles and challenges in single-cell data integration. *Nature biotechnology* **39**, 1202–1215 (2021).
- [29] Pasoli, E. *et al.* Accessible, curated metagenomic data through ExperimentHub. *Nature Methods* **14**, 1023–1024 (2017).
- [30] Rogers, A. B. *et al.* Holofood data portal: holo-omic datasets for analysing host–microbiota interactions in animal production. *Database (Oxford)* **2025**, baae112 (2025).
- [31] Lahti, L., Ernst, F. G. & Shetty, S. *microbiomeDataSets: Experiment Hub based microbiome datasets* (2025). R package version 1.16.0.
- [32] Gurbich, T. A. *et al.* MGnify genomes: A resource for biome-specific microbial genome catalogues. *Journal of molecular biology* **435**, 168016 (2023).
- [33] Borman, T., Sannikov, A. & Lahti, L. *HoloFoodR: R interface to EBI HoloFood resource* (2025). URL <https://github.com/EBI-Metagenomics/HoloFoodR>. R package version 1.3.0.
- [34] Borman, T., Allen, B. & Lahti, L. *MGnifyR: R interface to EBI MGnify metagenomics resource* (2025). URL <https://github.com/EBI-Metagenomics/MGnifyR>. R package version 1.5.0.
- [35] Marcos-Zambrano, L. J. *et al.* A toolbox of machine learning software to support microbiome analysis. *Frontiers in Microbiology* **14** (2023).
- [36] Rajilić-Stojanović, M. *et al.* Development and application of the human intestinal tract chip, a phylogenetic microarray: analysis of universally conserved phylo-types in the abundant microbiota of young and elderly adults. *Environmental Microbiology* **11**, 1736–1751 (2009).
- [37] Abakumov, S. *et al.* DynaMAP: Dynamic microbiome abundance profiling through high density optical mapping. *bioRxiv* (2024).
- [38] Blanco-Míguez, A. *et al.* Extending and improving metagenomic taxonomic profiling with uncharacterized species using metaphlan 4. *Nature biotechnology* **41**, 1633–1644 (2023).

- [39] Truong, D. T. *et al.* Metaphlan2 for enhanced metagenomic taxonomic profiling. *Nature methods* **12**, 902–903 (2015).
- [40] McDonald, D. *et al.* The biological observation matrix (BIOM) format or: how i learned to stop worrying and love the ome-ome. *Gigascience* **1**, 2047–217X–1–7 (2012).
- [41] McIver, L. J. *et al.* bioBakery: a meta’omic analysis environment. *Bioinformatics* **34**, 1235–1237 (2017).
- [42] Beber, M. E., Borry, M., Stamouli, S. & Yates, J. A. F. TAXPASTA: TAXonomic profile aggregation and STAndardisation. *Journal of Open Source Software* **8**, 5627 (2023).
- [43] Callahan, B. J. *et al.* DADA2: High-resolution sample inference from illumina amplicon data. *Nature Methods* **13**, 581–583 (2016).
- [44] Oliveira, F. S. *et al.* MicrobiomeDB: a systems biology platform for integrating, mining and analyzing microbiome experiments. *Nucleic Acids Research* **46**, D684–D691 (2018).
- [45] Zuur, A. F., Ieno, E. N. & Elphick, C. S. A protocol for data exploration to avoid common statistical problems. *Methods in Ecology and Evolution* **1**, 3–14 (2010).
- [46] Kullberg, R. F. J. *et al.* Association between butyrate-producing gut bacteria and the risk of infectious disease hospitalisation: results from two observational, population-based microbiome studies. *The Lancet Microbe* **5**, 100864 (2024).
- [47] Valles-Colomer, M. *et al.* The neuroactive potential of the human gut microbiota in quality of life and depression. *Nature Microbiology* **4**, 623–632 (2019).
- [48] Geistlinger, L. *et al.* BugSigDB captures patterns of differential abundance across a broad range of host-associated microbial signatures. *Nature Biotechnology* **42**, 790–802 (2024).
- [49] Gloor, G. B., Macklaim, J. M., Pawlowsky-Glahn, V. & Egozcue, J. J. Microbiome datasets are compositional: And this is not optional. *Frontiers in Microbiology* **8** (2017).
- [50] Martino, C. *et al.* A novel sparse compositional technique reveals microbial perturbations. *mSystems* **4**, e00016–19 (2019).
- [51] Silverman, J. D., Washburne, A. D., Mukherjee, S. & David, L. A. A phylogenetic transform enhances analysis of compositional microbiota data. *eLife* **6**, e21887 (2017).
- [52] McMurdie, P. J. & Holmes, S. Waste not, want not: Why rarefying microbiome data is inadmissible. *PLoS Computational Biology* **10**, 1–12 (2014).

- [53] Schloss, P. D. Waste not, want not: revisiting the analysis that called into question the practice of rarefaction. *mSphere* **9**, e00355–23 (2024).
- [54] Ruuskanen, M. O., Sommeria-Klein, G., Havulinna, A. S., Niiranen, T. J. & Lahti, L. Modelling spatial patterns in host-associated microbial communities. *Environmental Microbiology* **23**, 2374–2388 (2021).
- [55] Quince, C., Walker, A. W., Simpson, J. T., Loman, N. J. & Segata, N. Shotgun metagenomics, from sampling to analysis. *Nature Biotechnology* **35**, 833–844 (2017).
- [56] Bastiaanssen, T. F. S., Quinn, T. P. & Loughman, A. Bugs as features (part 1): concepts and foundations for the compositional data analysis of the microbiome–gut–brain axis. *Nature Mental Health* **1**, 930–938 (2023).
- [57] Faith, D. P. Conservation evaluation and phylogenetic diversity. *Biological Conservation* **61**, 1–10 (1992).
- [58] Armstrong, G. *et al.* Efficient computation of Faith’s phylogenetic diversity with applications in characterizing microbiomes. *Genome Research* **31**, 2131–2137 (2021).
- [59] Schloss, P. D. Rarefaction is currently the best approach to control for uneven sequencing effort in amplicon sequence analyses. *mSphere* **9**, e00354–23 (2024).
- [60] Willis, A. D. Rarefaction, alpha diversity, and statistics. *Frontiers in Microbiology* **10**, 2407 (2019).
- [61] Cassol, I., Ibañez, M. & Bustamante, J. P. Key features and guidelines for the application of microbial alpha diversity metrics. *Scientific Reports* **15**, 622 (2025).
- [62] Lozupone, C. & Knight, R. UniFrac: a new phylogenetic method for comparing microbial communities. *Applied Environmental Microbiology* **71**, 8228–8235 (2005).
- [63] Goodrich, J. *et al.* Conducting a microbiome study. *Cell* **158**, 250–262 (2014).
- [64] Holmes, I., Harris, K. & Quince, C. Dirichlet multinomial mixtures: Generative models for microbial metagenomics. *PLoS One* **7**, e30126 (2012).
- [65] DiGiulio, D. B. *et al.* Temporal and spatial variation of the human microbiota during pregnancy. *Proceedings of the National Academy of Sciences* **112**, 11060–11065 (2015).
- [66] Frioux, C. *et al.* Enterosignatures define common bacterial guilds in the human gut microbiome. *Cell Host & Microbe* **31**, 1111–1125.e6 (2023).

- [67] Lin, H. & Peddada, S. D. Multigroup analysis of compositions of microbiomes with covariate adjustments and repeated measures. *Nature Methods* **21**, 83–91 (2024).
- [68] Nickols, W. A. *et al.* MaAsLin 3: Refining and extending generalized multivariable linear models for meta-omic association discovery. *bioRxiv* (2024).
- [69] Clausen, D. S., Teichman, S. & Willis, A. D. Estimating fold changes from partially observed outcomes with applications in microbial metagenomics. *arXiv* (2024).
- [70] Pelto, J., Auranen, K., Kujala, J. & Lahti, L. Elementary methods provide more replicable results in microbial differential abundance analysis. *Briefings in Bioinformatics* **26**, bbaf130 (2025).
- [71] Jiang, H. *et al.* Multimedia: multimodal mediation analysis of microbiome data. *Microbiology Spectrum* **13**, e01131–24 (2025).
- [72] Klåvus, A. *et al.* “Notame”: Workflow for non-targeted LC–MS metabolic profiling. *Metabolites* **10** (2020).
- [73] Salehi, M. *et al.* Gender differences in global antimicrobial resistance. *npj Biofilms and Microbiomes* **11**, 79 (2025).
- [74] Lee, K. *et al.* Population-level impacts of antibiotic usage on the human gut microbiome. *Nature Communications* **14**, 1191 (2023).
- [75] Berg, G. *et al.* Microbiome definition re-visited: old concepts and new challenges. *Microbiome* **8**, 103 (2020).
- [76] Velten, B. *et al.* Identifying temporal and spatial patterns of variation from multimodal data using MEFISTO. *Nature Methods* **19**, 179–186 (2022).
- [77] Gao, Y. *et al.* miaSim: an R/Bioconductor package to easily simulate microbial community dynamics. *Methods in Ecology and Evolution* **14**, 1967–1980 (2023).
- [78] Salosensaari*, A. *et al.* Taxonomic signatures of cause-specific mortality risk in human gut microbiome. *Nature Communications* **12**, 2671 (2021).
- [79] Calle, M. L., Pujolassos, M. & Susin, A. coda4microbiome: compositional data analysis for microbiome cross-sectional and longitudinal studies. *BMC Bioinformatics* **24**, 82 (2023).
- [80] Nyholm, L. *et al.* Holo-Omics: Integrated host-microbiota multi-omics for basic and applied biological research. *iScience* **23**, 101414 (2020).
- [81] Qin, Y. *et al.* Combined effects of host genetics and diet on human gut microbiota and incident disease in a single population cohort. *Nature Genetics* **54**, 134–142

(2022).

- [82] Hintikka, J. *et al.* Xylo-oligosaccharides in prevention of hepatic steatosis and adipose tissue inflammation: Associating taxonomic and metabolomic patterns in fecal microbiomes with biclustering. *International Journal of Environmental Research and Public Health* **18** (2021).
- [83] Mallick, H. *et al.* An integrated Bayesian framework for multi-omics prediction and classification. *Statistics in Medicine* **43**, 983–1002 (2024).
- [84] Sherwani, M. K. *et al.* Multi-omics time-series analysis in microbiome research: a systematic review. *bioRxiv* (2025). URL <https://www.biorxiv.org/content/early/2025/07/06/2025.07.03.659054>.
- [85] Bastiaanssen, T. F. S., Quinn, T. P. & Cryan, J. F. Knowledge-based integration of multi-omic datasets with anansi: Annotation-based analysis of specific interactions. *arXiv* (2023).
- [86] Argelaguet, R. *et al.* MOFA+: a statistical framework for comprehensive integration of multi-modal single-cell data. *Genome Biology* **21**, 111 (2020).
- [87] Peschel, S., Müller, C. L., von Mutius, E., Boulesteix, A.-L. & Depner, M. NetCoMi: network construction and comparison for microbiome data in R. *Briefings in Bioinformatics* **22**, bbaa290 (2021).
- [88] Topçuoğlu, B. D. *et al.* mikropml: User-friendly R package for supervised machine learning pipelines. *Journal of Open Source Software* **6**, 3073 (2021).
- [89] Kou, Y., Xu, X., Zhu, Z., Dai, L. & Tan, Y. Microbe-set enrichment analysis facilitates functional interpretation of microbiome profiling data. *Scientific Reports* **10**, 21466 (2020).
- [90] Navarro, A. L. G., Koneva, N., Sánchez-Macián, A. & Hernández, J. A. A comprehensive guide to combining r and python code for data science, machine learning and reinforcement learning. *arXiv preprint arXiv:2407.14695* (2024).
- [91] Wootton, D. *The Invention of Science: A New History of the Scientific Revolution* First u.s. edition edn (Harper, an imprint of HarperCollins Publishers, New York, 2015). Paperback edition published December 13, 2016 by Harper Perennial.
- [92] Benedetti, G. *et al.* iSEETree: interactive explorer for hierarchical data. *Bioinformatics Advances* **5**, vbaf107 (2025).
- [93] Hutchison, W. J. *et al.* The tidyomics ecosystem: enhancing omic data analyses. *Nature Methods* **21**, 1166–1170 (2024).
- [94] Wilson, G. *et al.* Good enough practices in scientific computing. *PLoS Computational Biology* **13**, e1005510 (2017).

- [95] D’Elia, D. *et al.* Advancing microbiome research with machine learning: key findings from the ML4Microbiome COST action. *Frontiers in Microbiology* **14** (2023).
- [96] Joos, R. *et al.* Examining the healthy human microbiome concept. *Nature Reviews Microbiology* **23**, 192–205 (2024).
- [97] Cheng, L. *et al.* An additive gaussian process regression model for interpretable non-parametric analysis of longitudinal data. *Nature communications* **10**, 1798 (2019).
- [98] Awais, M. *et al.* Foundation models defining a new era in vision: a survey and outlook. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2025).
- [99] Mallick, H. *et al.* Predictive metabolomic profiling of microbial communities using amplicon or metagenomic sequences. *Nature communications* **10**, 3136 (2019).
- [100] Weiss, K., Khoshgoftaar, T. M. & Wang, D. A survey of transfer learning. *Journal of Big data* **3**, 9 (2016).
- [101] Zou, J. & Topol, E. J. The rise of agentic ai teammates in medicine. *The Lancet* **405**, 457 (2025).
- [102] Gao, S. *et al.* Empowering biomedical discovery with ai agents. *Cell* **187**, 6125–6151 (2024).

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Contributions

LL and TB initiated and coordinated the work. All authors contributed material, participated in manuscript preparation, and approved the final version.

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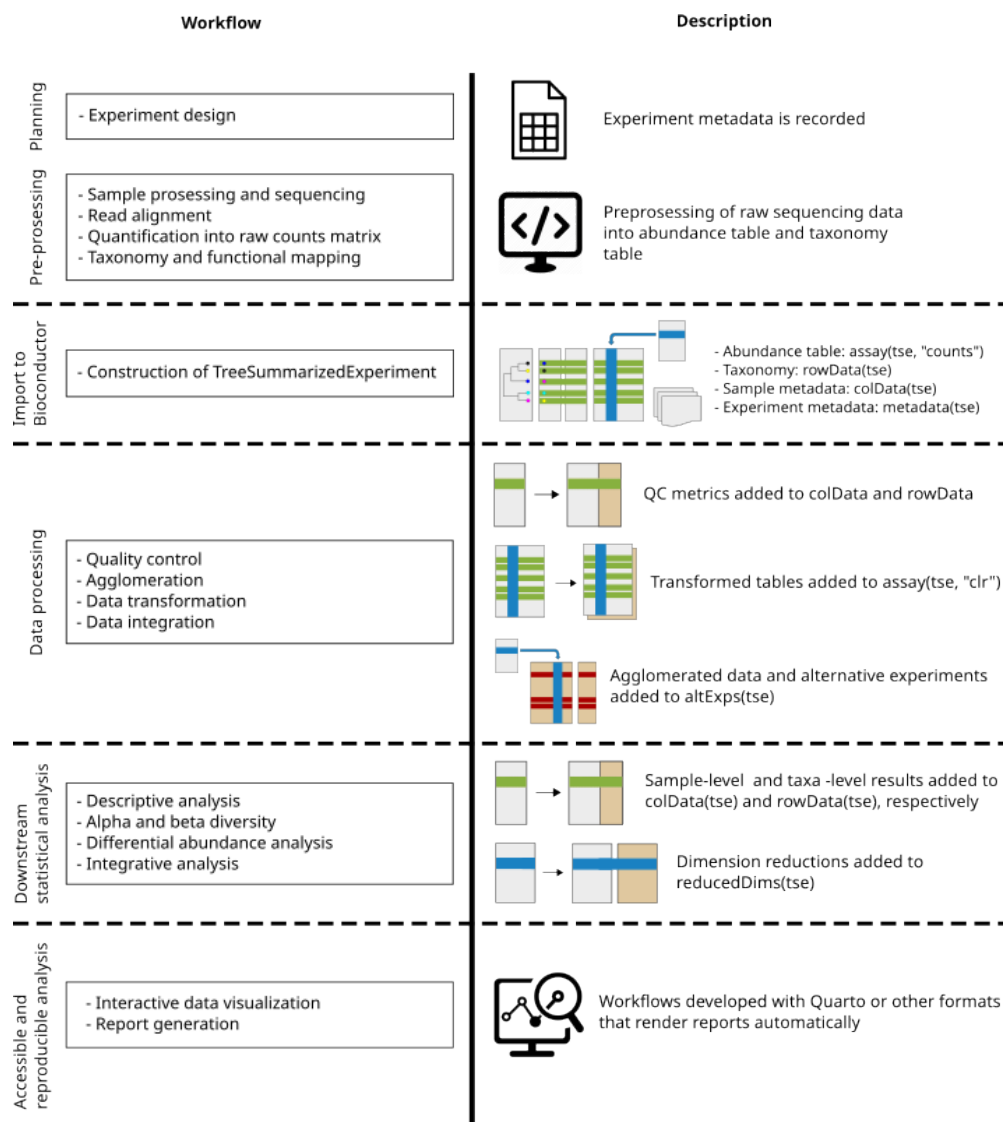


Figure 4: Examples of Bioconductor tools along the microbiome data science workflow. A typical microbiome workflow starts with taxonomic and functional annotation. The processed microbiome data can then be imported to Bioconductor data container (TreeSE), or integrated with other omics using the MAE container for downstream statistical analysis. (Figure adapted from ¹⁷)

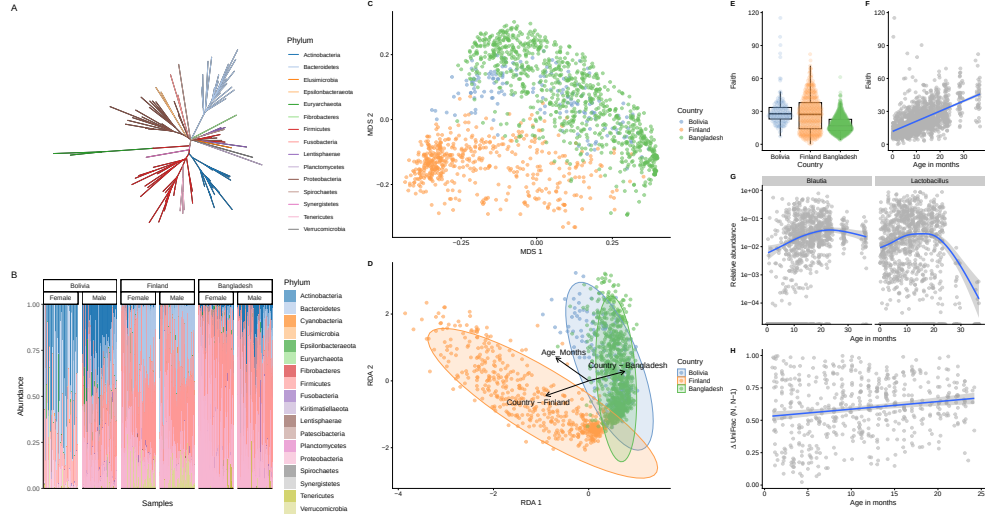


Figure 5: Common visualization methods. This set of visualizations present time series data on infants from three different countries. **Figure A** illustrates the phylogenetic relationships, while **Figure B** summarizes phyla abundances in the samples ordered based on age. Actinobacteria is more present in Bolivia and Bangladesh compared to Finland. **Figures C and D** display the results of Multidimensional Scaling (MDS) using Unifrac distances and Distance-based Redundancy Analysis (dbRDA) based on Aitchison distance, respectively. Both ordination methods shows clear pattern separating microbial profiles in different countries. dbRDA shows additionally association between microbial profile and age. **Figures E and F** depict Faith's phylogenetic diversity visualized in respect to country and age. **Figure G** visualizes Differential Abundance Analysis (DAA) results with scatter plot. Two genera exhibit association with age. **Figure H** illustrates Unifrac divergence for consecutive time points. Results indicate slow stabilization of microbial communities.